

Chapter – 3: Climates of India

May the rains be timely, may the Earth be lush with vegetation May this country be free from turmoil, may good people be fearless! – Subhāṣhita

Weather, Seasons and the Climate

- Climate – use in daily conversations – BUT – they really mean – weather – not climate
- Weather – experience every hour, day – raining, sunshine, wind, etc. – keeps changing
- Climate – pattern of weather – long period – several decades – varies – region to region
- Before climates – study seasons – occurs – earth's revolution around sun
- Each season – few months – recurs – every year
- Several seasons – spring, summer, monsoon, autumn, winter
- Seasons – related to climate, weather
- Weather changes with season – dry, hot – summers – humid, rainy – monsoon
- Pattern of seasons – closely connected – climate
- 4 main seasons – spring, summer, autumn, winter
- India – rainfall – specific period – rainy season – along with other 4
- Traditionally – India – many parts – year – divided – 6 seasons (*ritus*) – *Vasanta* (spring), *Grishma* (summer), *Varsha* (rainy), *Sharad* (autumn), *Hemanta* (pre-winter), *Shishir* (winter)
- Specific rituals, festivals – related to 6 *ritus*
- Obs. – world around us – notice – human, plant, animals – rhythm with seasons
- Crops, food, clothes – change – according to season
- Some plants – bloom at spring – some other – shed at autumn – some animals – develop thick fur in winters
- Climate – remains stable – long period of time
- BUT – scientists – recording changes – climates – recent decades – many changes – human actions

Types of Climates in India

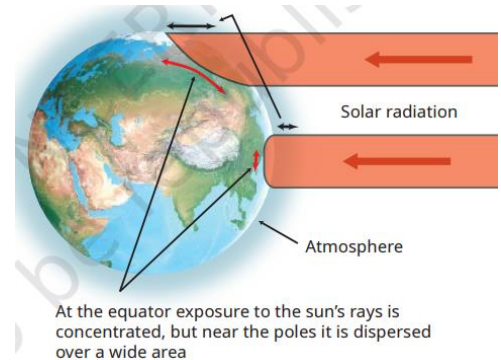
- India – land of diversity – equally true – climates
 - North – Himalayan mountains – alpine climate – cold, snowy winters – cool summers – word 'alpine' – from Alps – mountain range – Europe – thickest clothing – India
 - Lower Himalayas – many hilly areas – climate – temperate – winter – moderately cold – summers – not too hot – many 'hill stations' – frequented by people
 - Northern plains – climate – subtropical – very hot summers – cold winters – most wheat – grown here
 - West – Thar desert – arid climate – extremely hot days – cool nights – very little rainfall – people there – develop unique ways – collect, save water
 - Western coastal strip – heavy rainfall – monsoon months – tropical wet climate – favourable – rice, spices
 - Central Deccan Plateau – semi-arid climate – hot summers – mild winters – moderate rainfall – rainy season
 - Eastern India – southern peninsula – tropical climate – mild winter – distinct wet, dry periods – controlled by monsoon winds

Factors Determining the Climate

- Many factors – create diff. climates – some – general (planet level) – others – local, regional

Latitude

- Places – near equator – low latitudes – warmer – near poles – high latitudes – colder
- Reason – angle – sun's rays – hit particular area
- Equator – rays – perpendicular – all energy – focused – smaller area
- Polar regions – rays – inclined – energy – distributed – larger surface
- Also – they pass – more atm. – dissipates more – energy
- Result – polar regions – less heat than – equatorial zone
- India – Kanyakumari, Nicobar Islands – close to equator – warm – North – Srinagar – much cooler

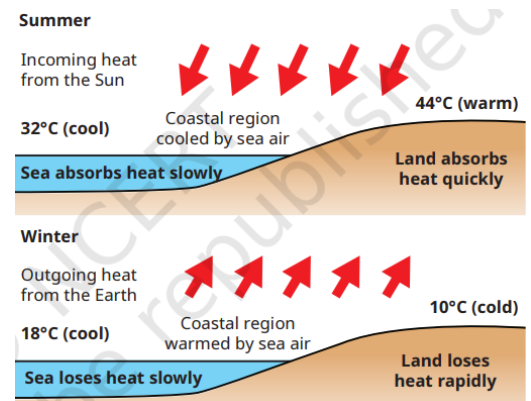


Altitude

- Hill stations – popular tourist place – cooler climate
- India – many stations – Munnar, Theni, Udhagamandalam (Ooty), Madikeri, Mahabaleshwar, Mount Abu, Shimla, Nainital, Darjeeling, Tawang, Shillong, etc.
- These places – higher alt. than – plains below – explain cooler temp.
- Keep things simple – alt. increases – temp. decreases –
 - Atm. pressure, air density – decreases – alt. increases – air – less dense – gets colder
 - Sun – heats the surface – farther away from surface – less hot air – Himalayas – so high – some peaks – maintain temp. – below freezing pt. – covered in snow

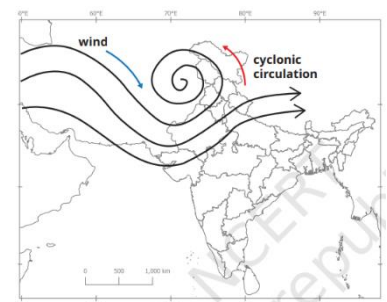
Proximity to the sea

- Temp. – coastal areas – do not vary much – summers – not too hot – winters – not too cold
- Reason – sea – act as – moderator – diagram – clears it up
- Result – these regions – more temp.
- Move inland – temp. – more extreme – summers – higher temp. – winters – lower temp.
- Mumbai, Nagpur – similar latitude –
 - Mumbai – near the sea – cooler summers (32⁰ C) – milder winters (18⁰ C)
 - Nagpur – away from coast – hotter summers (44⁰ C) – colder winters (10⁰ C)
- Mumbai's range – 14⁰ C – Nagpur's range – 34⁰ C



Winds

- Wind – move masses – warmer, cooler air
- States – Punjab, Haryana, Rajasthan, Madhya Pradesh – receive winds from west
- These winds – travelled from Arabia to Afghanistan – deserts – brings dry, hot air – severe heat waves



- Winters – cold winds from Himalayas – creep into Himalayan foothills – cold waves
- Winds – affect temp. – also humidity, precipitation
- Dry winds – from deserts – winds from sea – brings moisture – result – rainfall

Topography

- Topography – imp. role – determine climate
- Himalayas, Karakoram range – protect – Indian subcontinent – winds – cold deserts – Central Asia
- Flat topography – Thar desert – nothing to protect – hot, dry winds

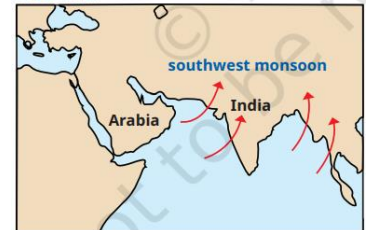
Putting it all together

- Climate – any region – decided together by – all factors
- Describing climate – patterns on temp., precipitation, wind conditions – 3 decades or more

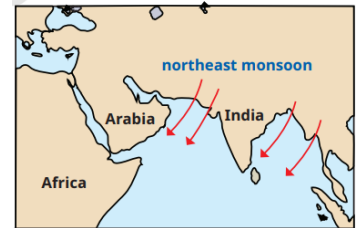
The Monsoons

- Monsoon season – central to life – India
- Monsoon months – rivers fill up – soil soaked – crops grow – life happens
- Word ‘monsoon’ – Arabic word ‘mausim’ – means – season – refers to – seasonal winds – large area – Indian Ocean – surrounding regions – Australia, Africa, South Asia
- Monsoon – yearly pattern – mechanism complex – based on fact – land heats up, cools down faster than ocean
- Monsoon – bring out – basic relation – temp., pressure, wind movement
- Summer begins – Asian landmass – heats up – low-pressure system
- Air – high to low pressure – wind – cooler, high-pressure ocean to hot land
- Ocean winds – carry moisture – condenses over – warmer land – falls – heavy monsoon rain
- Winters – pattern reverses – landmass cools down – rapidly – than ocean – high-pressure system
- Wind – opp. dir. – land to ocean – bring dry conditions – Asia
- India – monsoon winds – advance from southern tip – early June – move northward – several weeks – cover entire subcontinent – mid-July
- This progression – not smooth – Western Ghats – natural barrier – western slopes – heavy rainfall
- Eastern side – Deccan plateau – less rainfall – commonly called – summer / southwest monsoon
- Winter approaches – wind reverses – land to ocean
- These winds – dry – bring cold weather – south Indian
- Part of these winds – passing over Bay of Bengal – collect moisture – bring rainfall – east, south India – called as winter / northeast monsoon

A. Summer



B. Winter



Climate and our Lives

- Our lives – deeply connected – climate – depend on it
- Impact of climate – visible – local cultures – India – many festivals – seasonal, agricultural act.
- Climate – direct impact – economy
- Monsoon failure – poor rainfall – monsoon season –
 - Agriculture suffers – women walk long dist. for water

- Agricultural labourers migrate to cities – foods – costlier – fueling inflation
- Industrial act. – depends on – predictable weather – availability of water
- Easily detect – such connections – climate and socioeconomic conditions – get strained – climate disasters

Climates and Disasters

- India's diverse weather patterns – extreme conditions – cyclones, floods, landslides, etc.
- These events – affect lives – disrupt agriculture – damage infrastructure

Cyclones

- Every year – Indian coastline – specially eastern coast – several cyclones
- Past years – some of them – very destructive – result – loss of human, animal life – prop. damage – soil erosion
- IMD – Indian Meteorological Department – keeps track – provide info. – formation, evolution, landing place, etc.
- Some special situation – atm. pressure – sea area – lower than surroundings – low-pressure sys.
- This sys. – invitation – surrounding air – enter this sys. – air from sea – moves in – brings moisture, rain
- When – low-pressure sys. – intense – wind speeds – high – result – cyclone
- Winds – collect moisture – form clouds – rotate inwards – centre of depression
- This centre – cloudless – eye of cyclone



Floods

- Water overflows – normal dry land – flood
- Reason – heavy rainfall – huge run-off water – land cannot absorb – excess water – overflows – banks breached
- Floods – frequent – monsoon – states – Uttar Pradesh, Bihar, Kerala, Andhra Pradesh, Assam – more vulnerable
- Himalayan regions – floods – glacial lake overflows
- Glacial lakes – barrier of rocks, ice – hold water – melting glaciers
- Glaciers – melt too fast – heavy rainfall – built-up pressure – water breaks through – glacial burst – devastating consequences
- Many cities – flooding – heavy rainfall – reason – overburdened drainage sys. – poorly planned construction
- Urban surface – concrete, asphalt – water cannot be absorbed by ground

Landslides

- Landslide – sudden collapse – rock, soil, debris – triggered by – heavy rain, earthquake, volcano
- Landslides – common – hilly, mountainous areas – often during monsoon
- These regions – chances – increased – human activities – cutting trees – construction of too many buildings

Forest fires

- Uncontrolled fires spread rapidly – fueled by – dry climate conditions, droughts, high winds

- Human carelessness – another frequent reason
- Forest fires – common – states – large forested, grasslands areas – mountain ranges
- Fires – harm wildlife – degrade ecosystem – spoil air quality
- Consequences – both – environmental, economic

Climate Change

- Climate – change – significant, long-term changes – global, regional scale
- Involves – shift – temp., precipitation, weather events
- Past millennia – natural processes – climate changes
- After 19th Cent. – climate change – human activities – burning of fossil fuels – deforestation – industrial practices, production of wasteful consumption
- Earth's natural carbon cycle – CO₂, other gases – release gradually into atm. – trap heat
- This greenhouse effect – warms earth – support life
- Human activities – industry, transport, agriculture – released – huge amt. – greenhouse gases – few cent.
- Sudden increase – traps – extra heat – rapid global warming – disrupt climate patterns
- India – rising temp. – predictable – many regions
- Early 2025 – country's avg. temp. – 1⁰ – 3⁰ C – above normal – result – winter – shorter, milder
- This affects – agriculture, small-scale industries
- Understand relation – causes of climate change and disasters – prepare better
- Also support – environment-friendly practices – adaptation
- Govt. – all over world – promote measures – climate mitigation – cutting down – greenhouse emissions, planting trees – promoting sustainable lifestyle